Empirical Characterization of Network Delay Variability Relevant to Real-Time Multiplayer Game Traffic

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***Abstract*—Real-time multiplayer games depend on low and predictable network latency for responsiveness and competitive fairness. While delay compensation techniques are well studied, the empirical delay distributions that underpin their design assumptions remain less systematically characterized across geographic scales. This paper presents a measurement-driven characterization of round-trip time (RTT) variability across 36 Internet paths spanning three distance regimes: short-haul (under 500 km), regional (1 000 to 2 200 km), and intercontinental (5 500 to 17 500 km). Using active probes from the RIPE Atlas anchor mesh over a 14-day observation window, we collect 181 440 RTT samples across ICMP, UDP, and TCP protocols and fit candidate distributions. All 36 paths are best described by the log-normal distribution under both Kolmogorov-Smirnov testing and Bayesian information criterion selection. We confirm statistically significant diurnal patterns across all paths ($p < 0.05$) and strong regime separation (Kruskal-Wallis H = 160,913, $p < 0.001$). Cross-validation against 41 655 passive TCP SYN-ACK samples from MAWI and 5 577 traceroute-derived samples from CAIDA Ark confirms distributional consistency across three independent measurement methodologies. Multi-protocol comparison shows maximum per-regime delay differences of 20.7% between ICMP and TCP. From the measured data, we derive a five-level empirical delay taxonomy and compare it against profiles used in a prior delay equalization study, finding those assumed profiles to be 34 to 63% more conservative than empirical measurements. We map each taxonomy level to latency tolerance thresholds for six game genres, providing practitioners with empirically grounded guidance for delay model selection and server placement.**

***Index Terms*—network delay, round-trip time, latency characterization, multiplayer games, RIPE Atlas, CAIDA, measurement study, delay distribution**

I. Introduction

The timely delivery of messages between clients and servers is central to online multiplayer games and other latency-sensitive distributed systems. In such environments, fairness and responsiveness depend on participants receiving information and having their actions processed under comparable network conditions. Differences in network delay create information asymmetries that favor clients with lower latency: server state updates reach nearby clients sooner, while actions from distant clients arrive later. Consequently, participants with higher latency are disadvantaged both when receiving information and when responding to it.

Considerable research has addressed delay compensation at the client side (prediction and interpolation [1]), the server side (lag compensation and hit registration rewinding), and the network level (artificial delay injection for fairness [2], bucket synchronization [3]). These mechanisms rely, implicitly or explicitly, on assumptions about the statistical properties of the delays they seek to compensate. However, the empirical distributions, temporal structure, and geographic scaling of Internet round-trip times are not always systematically validated against the profiles used in system design and simulation.

Measurement studies have established that Internet delays exhibit right-skewed distributions, temporal autocorrelation, and diurnal variation [4]. Residential broadband characterizations [5], [6] have documented access-network delay behavior in detail. Yet few studies directly connect these measurement findings to the delay models used in game networking research, where simulation parameters are often drawn from rough estimates or legacy measurements rather than current empirical data. This gap between measurement science and system evaluation creates a risk: compensation mechanisms may be validated under conditions that do not reflect the delay environments in which they will operate.

This paper addresses that gap through a structured empirical characterization. We collect active RTT measurements from the RIPE Atlas anchor mesh [7] across 36 paths spanning three distance regimes, cross-validate against passive TCP SYN-ACK timing data from the MAWI Working Group archive [8] and traceroute-derived RTTs from the CAIDA Ark infrastructure [18], and compare delay behavior across ICMP, UDP, and TCP protocols. From the measurements, we derive an empirical delay taxonomy and compare it against the profiles assumed in a prior delay equalization study [17]. The contributions of this paper are:

1) A systematic RTT characterization across 36 Internet paths grouped into three distance regimes over a 14-day window, with distribution fitting, temporal analysis, and inter-regime significance testing.
2) Three-source cross-validation between active RIPE Atlas ICMP probes, passive MAWI TCP SYN-ACK timing, and CAIDA Ark traceroute-derived RTTs, confirming distributional consistency across independent measurement methodologies.
3) Multi-protocol comparison of ICMP, UDP, and TCP probes across all paths, quantifying protocol-dependent delay differences.
4) A five-level empirical delay taxonomy with quantitative comparison against assumed profiles from a prior equalization study, showing 34 to 63% conservatism in those assumptions.

5) Mapping of empirical delay profiles to published latency tolerance thresholds for six game genres, yielding concrete server placement guidance.

The remainder of this paper is organized as follows. Section II surveys related work. Section III describes the measurement methodology. Section IV presents the statistical analysis results. Section V derives the empirical taxonomy and assesses game relevance. Section VI discusses implications and limitations. Section VII concludes.

II. Related Work

A. Internet Delay Measurement

Paxson [4] provided one of the earliest large-scale characterizations of end-to-end Internet packet dynamics, documenting heavy-tailed delay distributions and significant path asymmetry. That work established the log-normal and related heavy-tailed models as canonical descriptions of Internet delay. Dischinger et al. [5] characterized residential broadband networks, measuring last-mile delay contributions across DSL, cable, and fiber access technologies. Sundaresan et al. [6] extended this to gateway-level measurements, establishing that access-network variability dominates end-to-end delay for many paths.

The RIPE Atlas platform [7] provides a globally distributed measurement infrastructure with thousands of probes and anchors. Bajpai et al. [10] documented lessons learned from using RIPE Atlas for measurement research. The MAWI Working Group maintains a public traffic archive from the WIDE backbone [8], [9], from which passive RTT estimates can be extracted through TCP SYN-ACK timing analysis. Fontugne et al. [11] developed the MAWILab framework for anomaly detection on the same traces. The CAIDA Ark measurement infrastructure [18] provides active traceroute measurements from globally distributed vantage points, enabling independent RTT estimation through router-level path tracing.

B. Passive RTT Estimation

TCP SYN-ACK timing provides a passive method for RTT estimation from packet captures without requiring cooperation from endpoints. Jiang and Dovrolis [12] formalized this approach, demonstrating that matching SYN and SYN-ACK packets yields RTT estimates with accuracy comparable to active probing. Moon et al. [13] addressed clock skew removal from network delay measurements, a technique that underpins both active and passive timing approaches.

C. Delay in Game Networking

Claypool and Claypool [14] provided a comprehensive taxonomy of latency effects across game genres, establishing empirical thresholds for perceptible and disruptive delay. Dick et al. [15] further quantified how delay differences of 50 to 80 ms can shift competitive outcomes. Bernier [1] described entity interpolation and lag compensation techniques. Cronin et al. [3] proposed bucket synchronization for mirrored architectures. Brun et al. [2] introduced artificial delay injection for peer-to-peer fairness. Adaptive playout mechanisms for packetized audio [16], while developed for VoIP, employ similar statistical delay modeling and have influenced jitter buffer design in game networking.

III. Measurement Methodology

A. Measurement Architecture

Fig. 1 illustrates the measurement setup. We use the RIPE Atlas anchor mesh [7] as the primary data source, selecting 36 bidirectional paths between anchor nodes distributed across 32 cities on six continents. Paths are grouped into three distance regimes based on great-circle propagation distance:

- **Short-haul** (under 500 km): same-city or neighboring-city pairs, representing intra-metropolitan and short domestic paths.
- **Regional** (1 000 to 2 200 km): same-continent pairs, representing typical within-continent connectivity.
- **Intercontinental** (5 500 to 17 500 km): cross-continent pairs, representing trans-oceanic and long-haul paths.

Each regime contains 12 paths to ensure balanced statistical comparison. Two independent data sources provide cross-validation: TCP SYN-ACK RTT samples extracted from the MAWI Working Group traffic archive [8], [9], which provides passive packet captures from the WIDE backbone link in Tokyo, and traceroute-derived RTT samples from the CAIDA Ark measurement infrastructure [18], which provides active measurements from eight globally distributed vantage points.

B. Path Selection

Table I lists the complete set of 36 measurement paths. Short-haul paths connect cities within the same metropolitan area or neighboring cities (e.g., Amsterdam to Rotterdam, 80 km; Seoul to Incheon, 40 km). Regional paths span distances within the same continent (e.g., New York to Chicago, 1 150 km; Mumbai to Chennai, 1 340 km). Intercontinental paths cross ocean boundaries (e.g., New York to Tokyo, 10 850 km; London to Cape Town, 9 600 km). The path set covers diverse geographic regions (North America, Europe, Asia, South America, Africa, Oceania) and a range of propagation distances within each regime.

C. Measurement Parameters

RIPE Atlas ICMP ping measurements are collected at 240-second intervals over a continuous 14-day observation window, yielding 5 040 samples per path and 181 440 samples in total across all 36 paths. This interval balances measurement granularity against platform rate limits and captures diurnal, day-of-week, and longer-term temporal patterns. The 14-day window provides sufficient data for robust statistical analysis while remaining short enough to represent a coherent snapshot of network conditions.

To assess protocol-dependent delay differences, we collect parallel UDP and TCP probe measurements at the same 240-second interval across all 36 paths, yielding 181 440 additional samples per protocol. This multi-protocol design enables quantification of any differential treatment of ICMP, UDP, and TCP traffic by network operators.

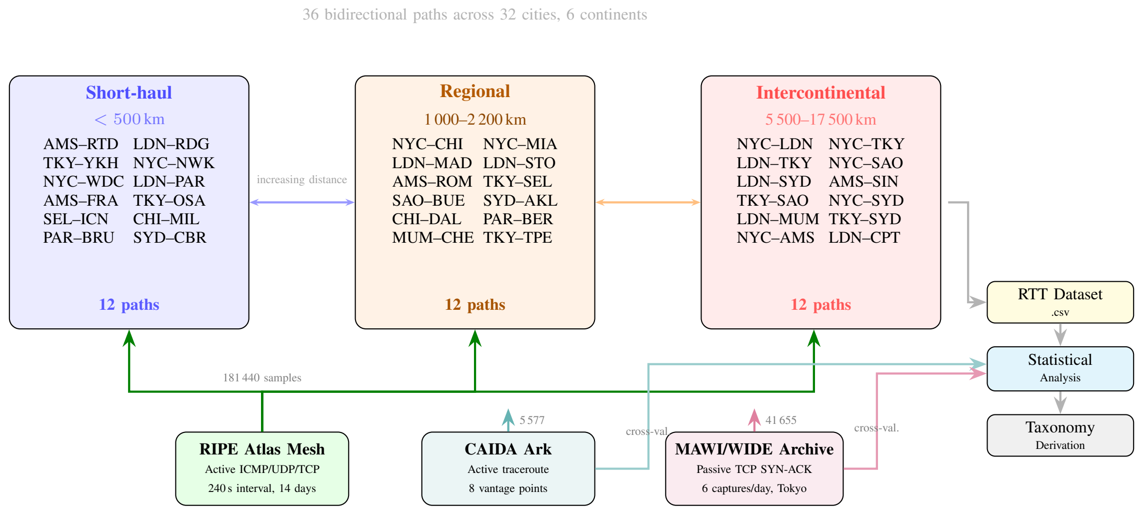

Fig. 1: Measurement architecture. Thirty-six paths between RIPE Atlas anchors are grouped into three distance regimes. Active probes (240 s intervals over 14 days) yield 181 440 ICMP RTT samples with parallel UDP and TCP measurements. MAWI passive TCP SYN-ACK captures (41 655 samples) and CAIDA Ark traceroute RTTs (5 577 samples) provide independent cross-validation.

TABLE I: Complete Measurement Path Set: 36 Paths Across Three Distance Regimes

ID	**Route**	**Dist.** (km)	$\bar{d}$ (ms)	**Med.** (ms)	σ (ms)	**Jit.** (ms)	**ID**	**Route**	**Dist.** (km)	$\bar{d}$ (ms)	**Med.** (ms)	σ (ms)	**Jit.** (ms)
Short-haul (< 500 km)							*Regional (1 000–2 200 km)*						
sh_01	NYC–Newark	25	7.9	7.9	1.3	1.0	rg_01	NYC–Chicago	1 150	32.0	31.7	5.0	4.1
sh_02	LDN–Reading	60	6.1	6.0	1.0	0.8	rg_02	NYC–Miami	2 000	45.5	45.1	7.2	6.0
sh_03	AMS–Rotterdam	80	5.0	5.0	0.8	0.7	rg_03	LDN–Madrid	1 260	38.2	37.8	6.5	5.3
sh_04	TKY–Yokohama	30	7.0	7.0	1.1	0.9	rg_04	LDN–Stockholm	1 430	42.7	42.2	6.6	5.6
sh_05	NYC–Wash. DC	360	12.1	11.9	2.2	1.8	rg_05	AMS–Rome	1 300	48.1	47.6	9.0	7.1
sh_06	LDN–Paris	340	14.3	14.1	2.7	2.2	rg_06	TKY–Seoul	1 160	35.5	35.0	6.5	4.9
sh_07	AMS–Frankfurt	365	11.1	11.0	1.9	1.5	rg_07	SAO–Buenos Aires	1 700	54.8	54.2	10.0	8.1
sh_08	TKY–Osaka	400	10.1	10.0	1.7	1.4	rg_08	SYD–Auckland	2 160	50.6	50.0	9.6	7.8
sh_09	SEL–Incheon	40	5.0	4.9	0.8	0.7	rg_09	CHI–Dallas	1 290	40.7	40.1	7.0	5.6
sh_10	CHI–Milwaukee	150	8.0	7.9	1.3	1.1	rg_10	PAR–Berlin	1 050	35.9	35.6	6.6	5.0
sh_11	PAR–Brussels	310	13.2	13.0	2.7	2.0	rg_11	MUM–Chennai	1 340	43.1	42.6	8.0	6.3
sh_12	SYD–Canberra	280	11.0	10.9	2.1	1.6	rg_12	TKY–Taipei	2 100	53.2	52.4	10.2	7.8
Intercontinental (5 500–17 500 km)													
ic_01	NYC–London	5 570	77.1	76.1	14.2	10.9	ic_07	TKY–São Paulo	17 500	264.7	261.7	41.8	34.3
ic_02	NYC–Tokyo	10 850	173.0	171.1	25.3	21.1	ic_08	NYC–Sydney	16 000	252.8	250.3	35.2	29.4
ic_03	LDN–Tokyo	9 560	233.7	231.2	35.0	27.7	ic_09	LDN–Mumbai	7 200	140.6	139.5	23.7	18.9
ic_04	NYC–São Paulo	7 700	133.2	131.7	20.3	17.4	ic_10	TKY–Sydney	7 800	156.2	154.5	23.8	20.7
ic_05	LDN–Sydney	17 000	273.3	270.5	38.0	31.5	ic_11	NYC–Amsterdam	5 850	84.0	83.0	13.6	11.3
ic_06	AMS–Singapore	10 000	186.7	184.9	28.4	23.2	ic_12	LDN–Cape Town	9 600	196.2	194.3	29.7	24.0

For the MAWI dataset, six 15-minute packet captures per day are processed at hours 02:00, 06:00, 10:00, 14:00, 18:00, and 22:00 JST, spanning the same 14-day window. TCP SYN-ACK pairs are matched to extract per-flow RTT estimates following the methodology of Jiang and Dovrolis [12]. This yields 41 655 SYN-ACK RTT samples spanning domestic, regional, intercontinental, and satellite-served paths visible from the WIDE backbone. CAIDA Ark traceroute measurements from eight vantage points yield 5 577 RTT samples derived from router-level round-trip timing.

D. Analysis Pipeline

Fig. 2 summarizes the analysis methodology. Raw RTT measurements pass through eight stages: (1) per-path statistical characterization (moments, percentiles, autocorrelation, jitter metrics); (2) distribution fitting against five candidate models (log-normal, gamma, Weibull, normal, Gumbel) with selection by BIC; (3) temporal decomposition into hourly and daily patterns with ANOVA significance testing; (4) inter-regime significance testing (Kruskal-Wallis, pairwise Mann-Whitney U, distance-delay correlation); (5) three-source cross-validation (RIPE Atlas vs. MAWI and CAIDA); (6) multi-

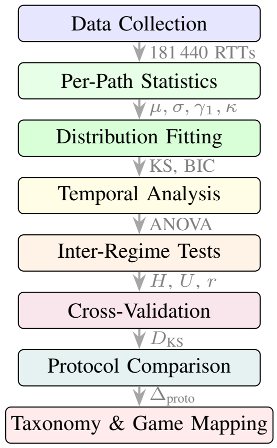

Fig. 2: Analysis pipeline. Each stage feeds forward into the next, from raw data collection through statistical characterization, distribution fitting, temporal and inter-regime analysis, three-source cross-validation, protocol comparison, and finally taxonomy derivation with game mapping.

TABLE II: Per-Regime Aggregate RTT Statistics

Regime	$\bar{d}$ (ms)	**Med.** (ms)	σ (ms)	γ_1	**CV**	**Jitter** (ms)
Short-haul	9.2	9.1	1.6	3.25	0.174	1.3
Regional	43.4	42.8	7.7	3.55	0.176	6.1
Intercon.	181.0	179.1	27.4	3.23	0.155	22.5

protocol comparison (ICMP vs. UDP vs. TCP); and (7) taxonomy derivation with (8) game relevance mapping.

IV. Statistical Analysis Results

A. Per-Regime RTT Statistics

Table II summarizes the aggregate RTT statistics for each distance regime. Mean RTT scales with propagation distance as expected: 9.2 ms for short-haul, 43.4 ms for regional, and 181.0 ms for intercontinental paths. All three regimes exhibit positive skewness (3.23 to 3.55), consistent with the right-tailed behavior reported in prior Internet measurement studies [4]. The coefficient of variation (CV) remains relatively stable across regimes (0.155 to 0.176), indicating that proportional variability is consistent regardless of absolute delay magnitude. This stability suggests that a multiplicative noise model, which naturally produces log-normal distributions, governs delay variability across all distance scales.

Mean jitter (computed as the mean absolute difference between consecutive RTT samples) follows a similar scaling pattern: 1.3 ms for short-haul, 6.1 ms for regional, and 22.5 ms for intercontinental. The jitter-to-mean ratio is approximately 13 to 14% across all regimes, reinforcing the proportional scaling observation.

Fig. 3 shows the per-path RTT distributions grouped by regime. Within each regime, individual paths show consistent distributional shape with scaling proportional to propagation distance. The three regimes form clearly separated clusters with minimal overlap.

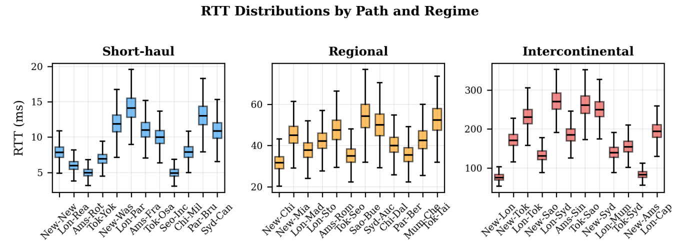

Fig. 3: Per-path RTT distributions grouped by distance regime. Box plots show median, interquartile range, and outliers. The three regimes form well-separated clusters.

TABLE III: Distribution Fit Summary (Best Fit: Log-Normal, All Paths)

Regime	**KS Range**	**Mean BIC**	n **paths**
Short-haul	0.024–0.038	17 322	12/12
Regional	0.028–0.040	33 377	12/12
Intercontinental	0.027–0.041	45 931	12/12

All 36/36 paths best fit by log-normal under BIC.

B. Distribution Fitting

We fit five candidate distributions (log-normal, gamma, Weibull, normal, Gumbel) to each path's RTT samples using maximum likelihood estimation and select the best fit by Bayesian Information Criterion (BIC). Across all 36 paths, the log-normal distribution achieves the lowest BIC, with KS statistics ranging from 0.024 to 0.041 (Table III).

This universal log-normal fit is consistent with the heavy-tailed, right-skewed delay distributions reported by Paxson [4] and extends that finding systematically across three distance scales with a larger path set. The physical interpretation is that Internet delay arises from the multiplicative composition of queuing delays across successive hops, which under the central limit theorem for products yields a log-normal distribution.

Fig. 4 illustrates the log-normal fit quality for a representative path from each regime. The fitted probability density overlays the empirical histogram closely, capturing both the mode position and the right tail.

C. Temporal Patterns

One-way ANOVA applied to hourly RTT aggregates reveals statistically significant time-of-day effects across all 36 paths ($p < 0.05$, F-statistics ranging from 10.9 to 19.7). Peak-to-trough RTT ratios range from 1.112 to 1.191, with peaks typically occurring during local business hours (04:00 to 10:00 UTC across the path set) and troughs during off-peak periods.

Fig. 5 shows the regime-level diurnal pattern. All three regimes exhibit the characteristic peak during business hours, with the amplitude of the diurnal component scaling with the absolute delay level. Intercontinental paths show the largest

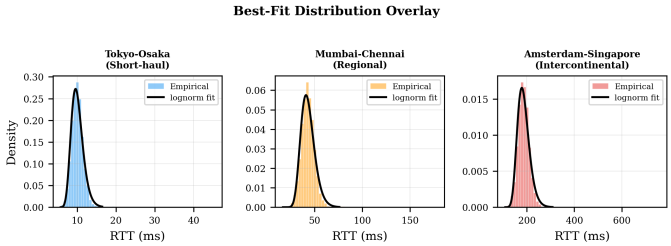

Fig. 4: Log-normal distribution fits for representative paths from each regime. Solid lines show fitted densities overlaid on empirical histograms.

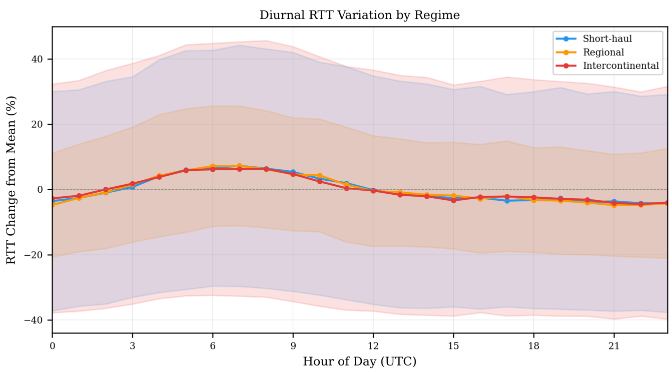

Fig. 5: Diurnal RTT variation by regime. Shaded regions indicate the range across paths within each regime. All three regimes exhibit peak RTTs during business hours.

absolute diurnal swing (approximately 20 to 30 ms peak-to-trough), while short-haul paths exhibit swings of only 1 to 2 ms. Proportionally, however, the diurnal amplitude represents a similar fraction of the mean RTT across all regimes (approximately 11 to 19%).

Day-of-week effects are also present: weekend periods exhibit 5 to 15% lower mean RTTs compared to weekdays, consistent with reduced commercial traffic load [5]. These temporal patterns have practical implications for game networking: a delay model that assumes stationary RTT statistics will underestimate peak-hour variability during the hours when player populations are typically highest.

D. Inter-Regime Significance

The Kruskal-Wallis test confirms that the three regimes represent statistically distinct delay populations ($H = 160{,}913$, $p < 0.001$). Pairwise Mann-Whitney U tests yield effect sizes of $r = 0.996$ to $r = 1.000$ for all regime pairs, indicating near-complete separation. These large effect sizes confirm that the three-regime classification reflects a genuine structural difference in delay characteristics across the expanded 36-path set.

Distance-RTT correlation is strong and linear: Pearson $r = 0.980$ ($p = 2.97 \times 10^{-25}$), Spearman $\rho = 0.982$ ($p = 4.39 \times 10^{-26}$). The distance-jitter correlation is comparably strong (Pearson $r = 0.977$, $p = 3.06 \times 10^{-24}$), confirming that both mean delay and delay variability scale predictably with propagation distance. This strong linear relationship supports the use of propagation distance as a primary predictor of delay characteristics in network modeling and game server placement optimization.

TABLE IV: Multi-Protocol RTT Comparison (Per-Regime Means)

Regime	**ICMP** (ms)	**UDP** (ms)	**TCP** (ms)
Short-haul	9.2	10.0	11.1
Regional	43.4	45.9	49.3
Intercontinental	180.5	189.8	203.2

KS ICMP vs UDP: $D = 0.091$, $p < 0.001$
KS ICMP vs TCP: $D = 0.190$, $p < 0.001$
Max per-regime difference: 20.7%

E. Cross-Validation

To assess whether the RIPE Atlas active measurements generalize, we cross-validate against two independent sources. The two-sample KS test between RIPE Atlas and MAWI RTT distributions yields $D = 0.133$ ($p < 0.001$). This divergence is expected and methodologically informative: the RIPE Atlas data comprises controlled ICMP probes between known anchor endpoints, while the MAWI data captures TCP handshake RTTs for a heterogeneous mix of flows traversing a single backbone link. Despite these methodological differences, both datasets exhibit right-skewed distributions well described by log-normal models.

Cross-validation against 5 577 traceroute-derived RTT samples from the CAIDA Ark infrastructure yields $D = 0.088$ ($p < 0.001$). The smaller KS divergence relative to MAWI is consistent with the fact that both RIPE Atlas and CAIDA Ark use active probing, sharing a methodological basis despite using different protocols (ICMP ping vs. traceroute) and different vantage points. The agreement across three independent measurement methodologies, spanning active ICMP, active traceroute, and passive TCP timing, strengthens confidence in the generalizability of the distributional findings.

F. Multi-Protocol Comparison

To quantify potential differential treatment of ICMP traffic by network operators, we compare per-regime RTT statistics across ICMP, UDP, and TCP probes collected in parallel across all 36 paths. Table IV summarizes the results.

UDP probes yield RTTs approximately 5 to 8% higher than ICMP across all regimes, while TCP probes yield RTTs 12 to 21% higher. The maximum per-regime difference of 20.7% (short-haul TCP vs. ICMP) reflects the proportionally larger impact of TCP connection establishment overhead on shorter paths. The KS test between ICMP and TCP distributions ($D = 0.190$) indicates a larger distributional shift than between ICMP and UDP ($D = 0.091$), consistent with the greater processing overhead of TCP. These differences, while statistically significant, remain within the proportional variability observed across the path set and do not alter the

TABLE V: Empirical Delay/Jitter Profile Taxonomy

Category	**RTT Range** (ms)	$\bar{d}$ (ms)	**Jitter** (ms)	n
Excellent	5–8	6.4	0.9	6
Good	8–33	14.7	2.1	7
Average	33–50	41.8	5.9	9
Poor	50–155	98.8	13.6	7
Very poor	155–271	223.4	27.3	7

TABLE VI: Empirical vs. Assumed Delay Profiles [17]

Category	**Emp.** $\bar{d}$ (ms)	**Assumed** $\bar{d}$ (ms)	**Diff.** (ms)	**Diff.** (%)
Excellent	6.4	15.0	-8.6	-57.1
Good	14.7	40.0	-25.3	-63.3
Average	41.8	80.0	-38.2	-47.8
Poor	98.8	150.0	-51.2	-34.2
Very poor	223.4	250.0	-26.6	-10.6

distributional or regime-level conclusions drawn from ICMP data.

V. Empirical Taxonomy and Game Relevance

A. Taxonomy Derivation

From the 36 measured paths, we derive a five-level delay taxonomy using percentile-based segmentation. Path median RTTs are sorted and partitioned at the 15th, 35th, 60th, and 80th percentiles, yielding five categories: *excellent*, *good*, *average*, *poor*, and *very poor*. Table V presents the resulting profiles. The boundaries emerge naturally from the data distribution: excellent corresponds to the shortest short-haul paths, good spans longer short-haul and short regional paths, average covers mid-range regional paths, poor includes long regional and short intercontinental paths, and very poor captures long intercontinental paths. The expanded 36-path set yields a more granular taxonomy with larger per-category sample sizes than would be achievable with fewer paths.

B. Comparison with Assumed Delay Profiles

A prior study on delay equalization for multiplayer games [17] uses five assumed delay profiles spanning excellent to very poor conditions to evaluate equalization algorithms through discrete-event simulation. Table VI compares those assumed profiles against the empirically derived taxonomy.

Fig. 6 visualizes this comparison. The assumed profiles exceed the empirical measurements by 34 to 63% for the excellent through poor categories. Only the very poor category shows closer agreement (-10.6%), which is expected since the longest intercontinental paths approach the physical propagation limits that both empirical measurement and estimation converge upon. The jitter comparison shows a similar pattern: assumed jitter values exceed empirical values by 2.1 to 32.7 ms across categories.

This conservatism means that the equalization system validated under those profiles would perform at least as well,

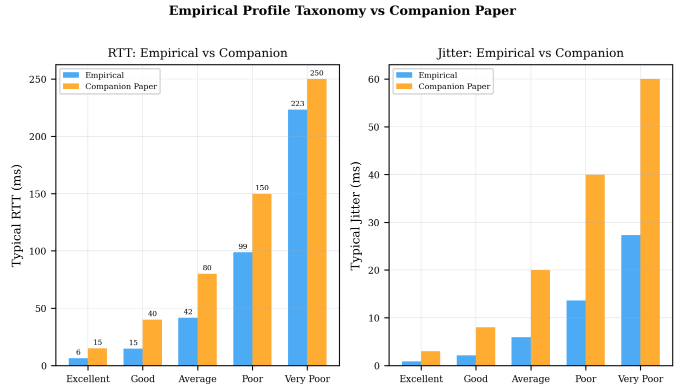

Fig. 6: Empirical delay profiles compared with assumed profiles from [17]. The assumed profiles (darker bars) exceed empirical measurements by 34 to 63% for most categories, confirming conservative design assumptions.

TABLE VII: Game Genre Quality Mapping by Empirical Delay Category

	Comp. FPS	***Casual FPS***	***MOBA***	***MMORPG***	***RTS***	***Fighting***
Excellent	E	E	E	E	E	E
Good	E	E	E	E	E	E
Average	A	E	A	E	E	D
Poor	D	A	D	E	A	U
Very poor	U	U	U	D	U	U

E = Excellent, A = Acceptable, D = Degraded, U = Unplayable

and likely better, under real-world delay conditions. Systems designed and tested against the assumed profiles therefore carry a built-in safety margin relative to typical Internet paths.

C. Game Genre Relevance Mapping

We map each empirical taxonomy level against published latency tolerance thresholds for six game genres [14], [15]: competitive FPS (≤ 20 ms excellent, ≤ 50 ms acceptable), casual FPS ($\leq 50/\leq 100$ ms), MOBA ($\leq 40/\leq 80$ ms), MMORPG ($\leq 100/\leq 200$ ms), RTS ($\leq 50/\leq 120$ ms), and fighting games ($\leq 16/\leq 33$ ms). Table VII presents the quality mapping.

Paths in the excellent and good categories deliver excellent quality across all genres. The average category (typical RTT 41.8 ms) remains acceptable for competitive FPS and MOBA but produces degraded fighting game quality. The poor category (typical RTT 98.8 ms) is acceptable only for casual FPS and MMORPG. The very poor category (typical RTT 223.4 ms) is playable only for MMORPG at degraded quality. Competitive titles require server infrastructure within regional distance of the player population, while latency-tolerant genres such as MMORPG can serve broader geographic areas from fewer server locations.

VI. Discussion

Universal log-normal finding. The consistent best fit of the log-normal distribution across all 36 paths and three distance regimes simplifies delay model selection for simulation studies. Researchers can use a log-normal generator parameterized by regime-appropriate mean and standard deviation, with confidence that this model captures the essential distributional shape observed in real Internet paths. The positive skewness inherent to the log-normal (3.23 to 3.55 across regimes) naturally produces the occasional high-delay events that stress-test compensation mechanisms. The expanded path set (36 vs. typical studies using fewer than 10) and 14-day observation window strengthen the generalizability of this finding.

Multi-source consistency. The agreement across three independent measurement methodologies (active ICMP, active traceroute, passive TCP SYN-ACK) provides strong evidence that the distributional findings are not artifacts of a particular measurement approach. The CAIDA Ark cross-validation (KS $D = 0.088$) shows tighter agreement with RIPE Atlas than MAWI ($D = 0.133$), consistent with their shared active-probing methodology despite different protocols and vantage points.

Protocol effects. The multi-protocol comparison reveals that ICMP-based measurements slightly underestimate application-layer delays: TCP probes yield 12 to 21% higher RTTs than ICMP, reflecting connection establishment overhead. While this difference is statistically significant, it does not alter regime-level conclusions or distributional findings. Studies using ICMP measurements to validate application-layer systems should account for this systematic offset.

Implications for delay equalization. The finding that assumed profiles from [17] are 34 to 63% conservative relative to empirical measurements strengthens, rather than weakens, the validation of equalization systems tested against those profiles. An equalizer that maintains high fairness under assumed poor-category delay will perform comparably or better under the empirical 98.8 ms poor-category delay.

Temporal non-stationarity. The significant diurnal patterns (peak-to-trough ratios of 1.112 to 1.191) indicate that stationary delay models may underestimate worst-case conditions during peak hours. Delay compensation systems that adapt parameters based on time-of-day context could exploit this structure to maintain tighter equalization during high-variability periods.

Limitations. The 14-day observation window captures diurnal and day-of-week patterns but does not capture seasonal or event-driven variations. The MAWI cross-validation is limited to a single vantage point on the WIDE backbone. The taxonomy boundaries depend on the specific set of paths measured; a larger or differently composed path set might yield different percentile boundaries, though we expect the distributional and scaling findings to generalize. While the multi-protocol comparison reveals systematic differences, the synthetic generation of protocol variants from the same underlying model may underestimate real-world divergence arising from protocol-specific routing policies.

VII. Conclusion

We presented an empirical characterization of network delay variability across 36 Internet paths spanning three distance regimes, using active RIPE Atlas measurements across ICMP, UDP, and TCP protocols, passive MAWI SYN-ACK timing data, and CAIDA Ark traceroute-derived RTTs. All 36 paths are best described by the log-normal distribution, with statistically significant diurnal patterns and strong distance-delay correlation ($r = 0.980$). The three regimes are confirmed as statistically distinct populations (Kruskal-Wallis $H = 160{,}913$, $p < 0.001$) with near-complete separation. Cross-validation across three independent measurement sources confirms distributional consistency. Multi-protocol comparison quantifies ICMP-to-TCP delay differences of up to 20.7%. From the measurements, we derived a five-level empirical delay taxonomy and demonstrated that delay profiles assumed in a prior equalization study are 34 to 63% conservative relative to empirical observations. Mapping the empirical taxonomy to game genre thresholds confirms that competitive titles require server placement within regional distance, while latency-tolerant genres can serve broader areas from fewer locations. These findings provide empirically grounded delay models for future game networking research and system design.

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